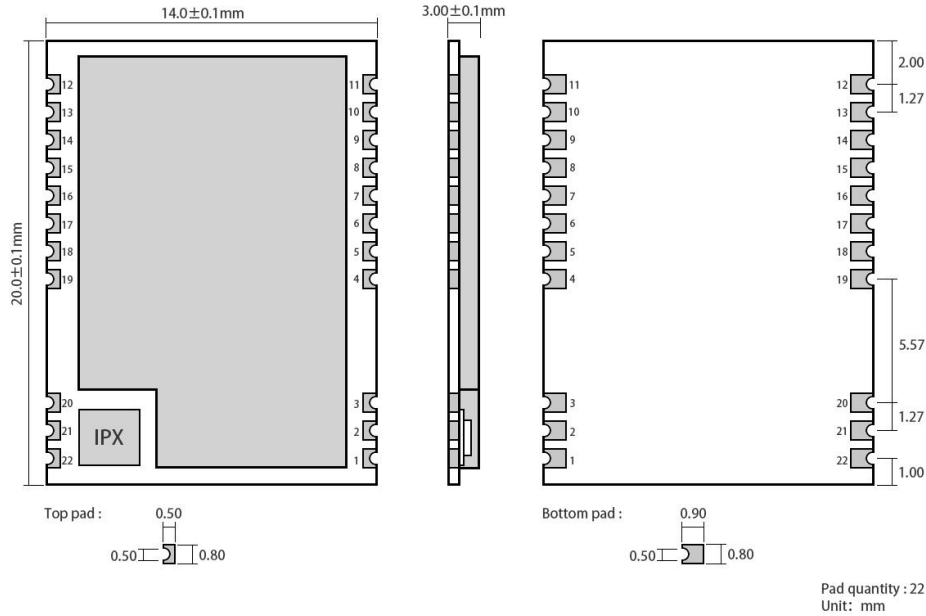


3 Size and pin definition



Pin No.	Item	Direction	Description
1	GND		Ground, connect to power reference ground
2	GND		Ground, connect to power reference ground
3	GND		Ground, connect to power reference ground
4	GND		Ground, connect to power reference ground
5	DIO3		Configurable universal IO port (see llcc68 manual for details)
6	RXEN	input	RF switch receive control pin, connect external MCU IO, high level effective
7	TXEN	input	RF switch emission control pin, connected to external MCU IO or dio2, high level effective
8	DIO2	Input / output	Configurable universal IO port (see llcc68 manual for details)
9	VCC		Power supply, range 1.8V ~ 3.7V (it is recommended to add ceramic filter capacitor externally)
10	GND		Ground, connect to power reference ground
11	GND		Ground, connect to power reference ground
12	GND		Ground, connect to power reference ground
13	DIO1	Input / output	Configurable universal IO port (see llcc68 manual for details)
14	BUSY	output	For status indication (see llcc68 manual for details)

15	NRST	input	Chip reset trigger input pin, low level effective
16	MISO	output	SPI data output pin
17	MOSI	input	SPI data input pin
18	SCK	input	SPI clock input pin
19	NSS	input	The module chip selection pin is used to start a SPI communication
20	GND		Ground, connect to power reference ground
21	ANT		RF interface, stamp hole
22	GND		Ground, connect to power reference ground

For the pin definition, software driver and communication protocol of the module, please refer to the official llcc68 datasheet of Semtech

4 Basic operation

4.1 Hardware design

- It is recommended to use a DC stabilized power supply. The power supply ripple factor is as small as possible and the module needs to be reliably grounded;
- Please pay attention to the correct connection of the positive and negative poles of the power supply, reverse connection may cause permanent damage to the module;
- Please check the power supply to ensure that between the recommended supply voltage, if exceeding the maximum, the module will be permanently damaged;
- Please check the stability of the power supply. Voltage can not fluctuate greatly and frequently;
- When designing the power supply circuit for the module, it is often recommended to reserve more than 30% of the margin, so the whole machine is beneficial for long-term stable operation;
- The module should be as far away as possible from the power supply, transformers, high-frequency wiring and other parts with large electromagnetic interference;
- Bottom LayerHigh-frequency digital routing, high-frequency analog routing, and power routing must be avoided under the module. If it is necessary to pass through the module, assume that the module is soldered to the Top Layer, and the copper is spread on the Top Layer of the module contact part(well grounded), it must be close to the digital part of the module and routed in the Bottom Layer;
- Assuming the module is soldered or placed over the Top Layer, it is wrong to randomly route over the Bottom Layer or other layers, which will affect the module's spurs and receiving sensitivity to varying degrees;
- It is assumed that there are devices with large electromagnetic interference around the module that will greatly affect the performance. It is recommended to keep them away from the module according to the strength of the interference. If necessary, appropriate isolation and shielding can be done;
- Assume that there are traces with large electromagnetic interference (high-frequency digital, high-frequency analog, power traces) around the module that will greatly affect the performance of the module. It is recommended to stay away from the module according to the strength of the interference.If necessary, appropriate isolation and shielding can be done;
- If the communication line uses a 5V level, a 1k-5.1k resistor must be connected in series (not recommended, there is still a risk of damage);
- Try to stay away from some physical layers such as TTL protocol at 2.4GHz , for example: USB3.0;